GradMap

Team Ruby

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Problem

Students struggle to create efficient class schedules due to complex degree requirements, prerequisite chains, and scheduling conflicts.

Solution

GradMap automatically generates personalized, conflict-free class schedules based on degree requirements, availability, and student preferences.

Problem Characteristics

User Limitations:

- Errors when balancing course loads and prerequisites

Scheduling Challenges:

- conflicts between class times and personal commitments

System Gaps:

- lack of real-time integration with course availability
- heavy reliance on manual advisor review

Impact / Why It Matters

Impact: GradMap helps students plan courses more accurately and avoid scheduling conflicts.

Why it matters: This keeps students on track to graduate on time and reduces advising errors.

Solution Characteristics

Smart Scheduling:

- Automated schedule generation based on requirements and availability
- Built-in conflict detection and resolution

Personalization:

- Customized plans with balanced course load recommendations

User Experience:

- Easy navigation and clear degree progress tracking

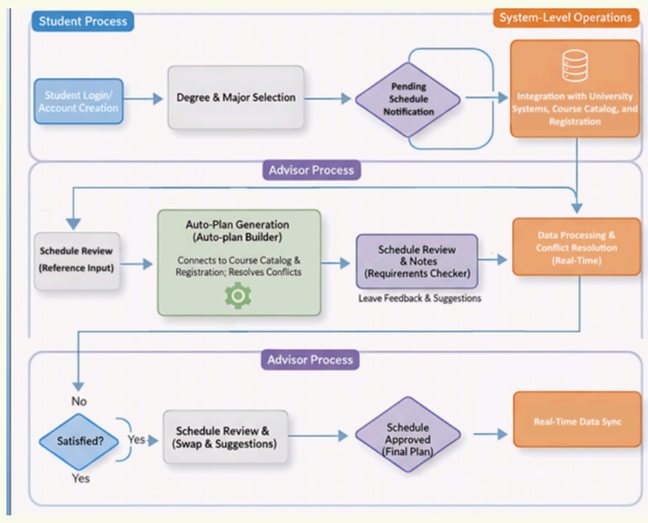
Collaboration:

- Centralizes student and advisor communication

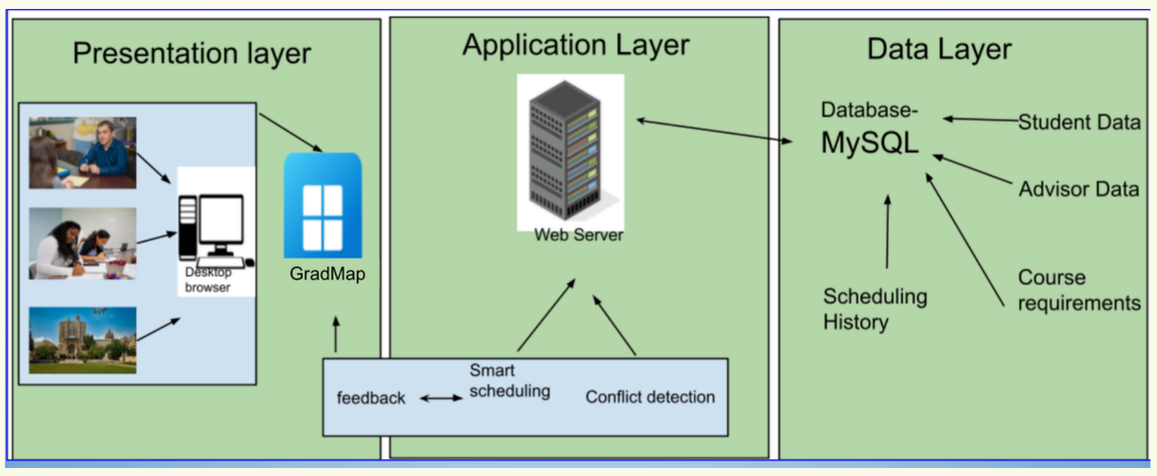
Innovative Features vs. Competition

Features	GradMap	Degreeworks	Internal University Registration Systems
Automated Smart Scheduling			
Conflict Detection		✔	✔
Load-Balancing Recommendations			
Streamlined Workflow		✔	
Constraint Blocks for Students			✔
Advisor Review		✔	✔
Requirements Checker		✔	✔

MFCD for RWP



Prototype MFCD



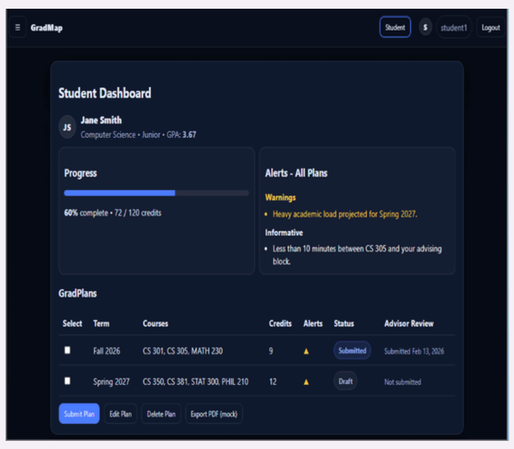
Prototype Features

Features	Real-World Product (RWP)	Prototype
Automated Degree Requirement Checker	Automatically updates degree requirements across different programs.	Basic rule-checking with manually entered program requirements.
Auto-Plan Builder	Generates complete, conflict-free academic plans.	Creates simplified schedules using a limited set of restraints.
Smart Suggestions	Delivers course alternatives, elective recommendations, and optimized pathways.	Simple rule-based suggestions to highlight the idea without full AI.
Student Profile Integration	Fully integrates with the university system to import data. (Coursework, degree progress, etc.)	Uses manual input for student data due to lack of SIS access.
Advisor Dashboard	Provides advisors with review tools, analytics, and intervention insights.	Offers a basic approval and commenting interface.
Real-Time Conflict Detection	Instantly flags time conflicts, prerequisite issues, and policy issues.	Detects only time conflicts an basic prerequisites.
Collaboration Tools	Supports real-time communication and plan editing between students and advisors.	A simple comment system wi real-time syncing or messagi
Authentication and Role Management	Provides secure logins for students, advisors, and administrators.	Basic separate login pages tc differentiate student and advi access.

UI/UX Screens

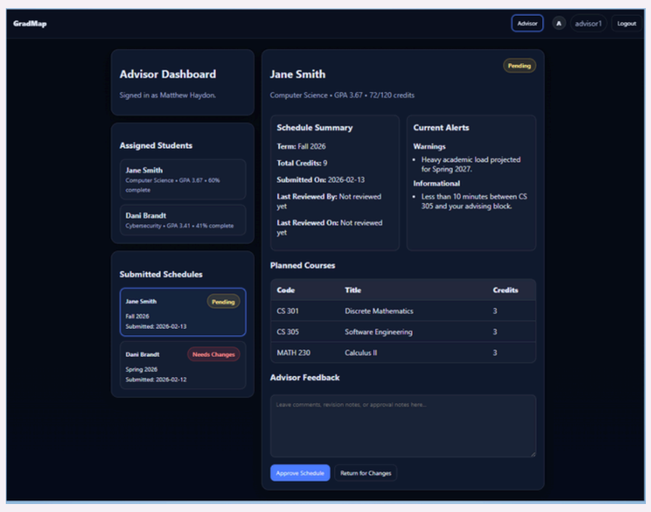
Student Dashboard

Displays student progress, alerts, and generated schedules in one centralized view



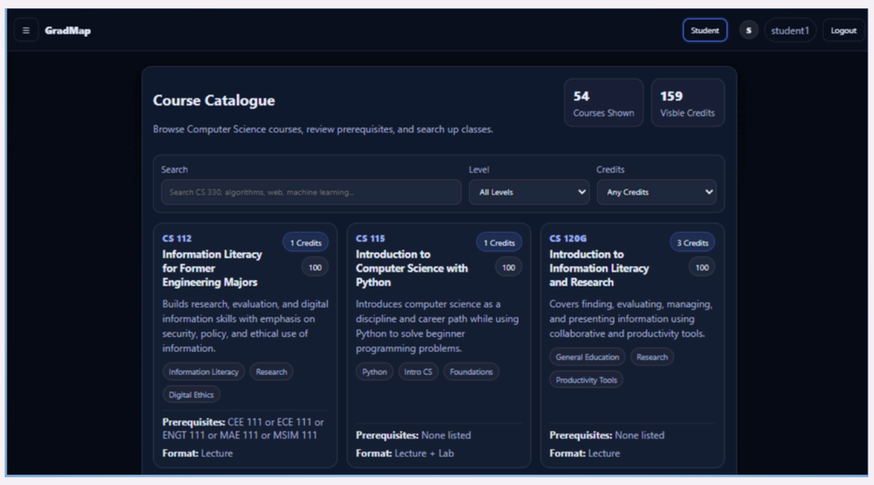
Advisor Dashboard

Allows advisors to review student schedules, monitor progress, and provide feedback or approvals



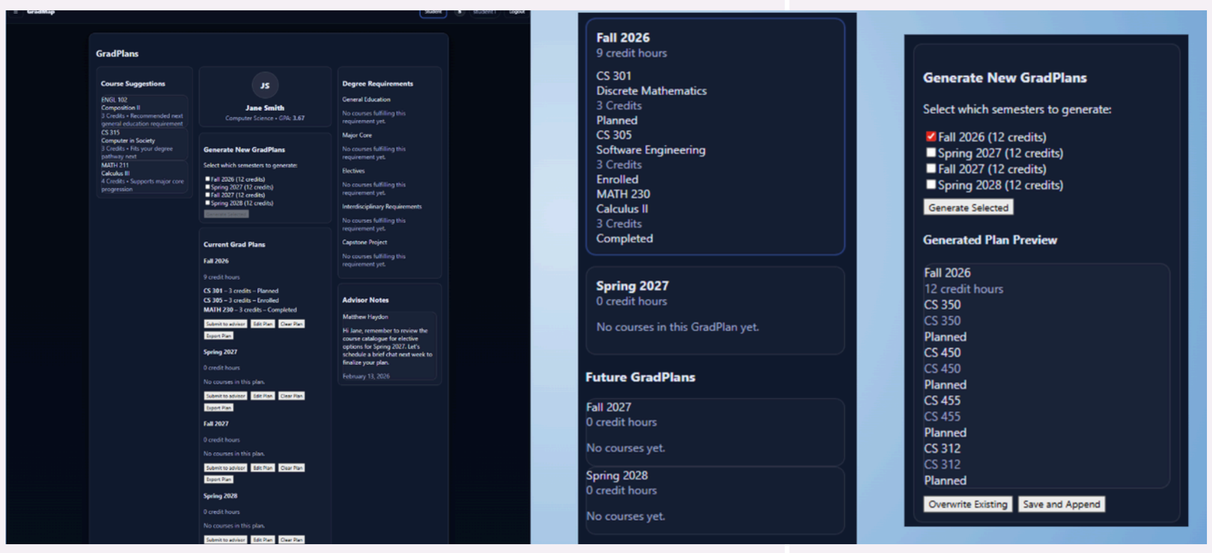
Course Catalogue

Allows students to generate, view, and manage personalized semester by semester course plans.



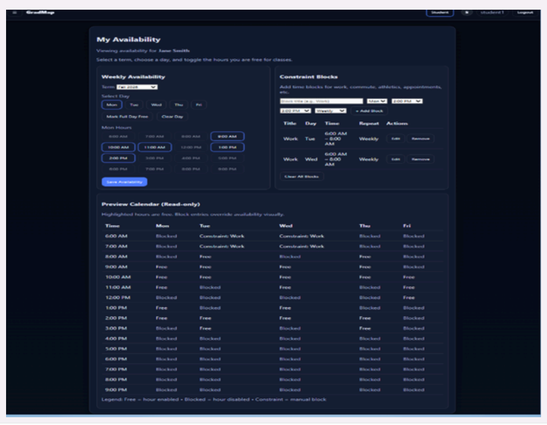
GradPlans

Allows students to generate, view, and manage personalized semester by semester course plans.



My Availability Dashboard

Lets students input their availability and constraints to generate schedules that fit their personal commitments.



See Team Ruby's full website: https://chat.google.com/room/AAQAhyYZmZ0/x14TwOiBEdY/x14TwOiBEdY?cls=10